# Urban Flooding and Adaptation

Allan Hsiao
Stanford

Dev Patel
Harvard

March 6, 2026

# This paper

- Spatial sorting by income
  - Compensating differentials for environmental risk
- Adaptation policy faces a trade-off
  - Protecting risky areas reduces inequality in exposure
  - But also reduces housing affordability
- Today: theory and data from Indian cities

# Residential choice

$$v_{ij} = -\alpha_j p_i - \beta_j e_i$$

- Locations $i \in \{$High risk, Low risk$\}$
- Groups $j \in \{$Poor, Rich$\}$, where $\alpha_P > \alpha_R$, $\beta_P > \beta_R$

- Location utility $v_{ij}$
  - Housing prices $p_i$
  - Flood exposure $e_i = n_i - g_i$
  - Natural $n_i$, where $n_H > n_L$
  - Infrastructure $g_i$

# Equilibrium prices

$$D_i(\hat{p}) = \sum_j \pi_{ij}(\hat{p}), \quad S_i = 1$$

- Demand from logit choice probabilities $\pi_{ij}$
- Supply from fixed housing stock $S_i$

# Compensating differentials

$$\underbrace{\Delta \hat{p}}_{\hat{p}_H - \hat{p}_L} = -\bar{\gamma} \Delta e, \quad \bar{\gamma} = \frac{\beta_P + \beta_R}{\alpha_P + \alpha_R}$$

- Higher risk in $H$, lower price in $H$
- Market clears on aggregate $\bar{\gamma}$ (WTP to avoid flooding)

# Sorting by income

$$c = \log\left(\frac{\pi_{HP}}{\pi_{LP}}\right), \quad \frac{\mathrm{d}\hat{c}}{\mathrm{d}\Delta e} = \alpha_P(\bar{\gamma} - \gamma_P)$$

- Sorting: concentration $c$ of the poor in $H$, given exposure gap $\Delta e$
  - The rich crowd out the poor in pursuit of higher ground (Hsiao 2024)
  - Worsens inequality in exposure

- For $\gamma_j = \beta_j / \alpha_j$, sorting iff

$$\boxed{\bar{\gamma} > \gamma_P} \quad \text{and} \quad \bar{\gamma} < \gamma_R$$

# Welfare by income

$$\frac{\mathrm{d}\hat{U}_j}{\mathrm{d}\Delta e} = (\bar{\gamma} - \gamma_j)\pi_{Hj}$$

- Under sorting, $\boxed{\bar{\gamma} > \gamma_P}$ and $\bar{\gamma} < \gamma_R$
  - The poor gain from exposure differences; the rich lose
  - Reduces inequality in welfare

- Market clears on aggregate $\bar{\gamma}$ (WTP)
  - For the poor, over-compensation through housing prices

# Adaptation policy faces a trade-off

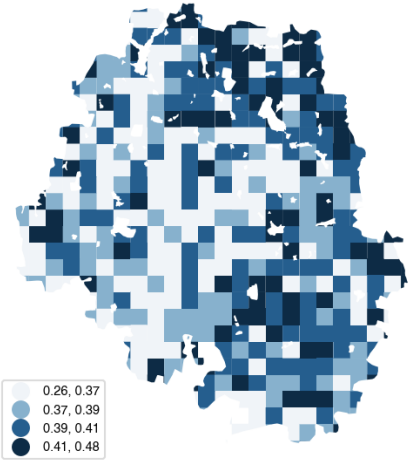
- In-situ and ex-situ adaptation interact
  - Policy: government infrastructure (in situ)
  - Response: migration and sorting (ex situ)
- Exposure-based allocation protects $H$ ($n_H > n_L$)
  - Exposure gap $\downarrow$, aggregate losses $\downarrow$
  - But have gentrification and uniformly expensive housing
- Price-based allocation protects $L$ ($\hat{p}_L > \hat{p}_H$)
  - Exposure gap $\uparrow$, aggregate losses $\uparrow$
  - But preserves pockets of affordable housing

# New data on flooding in Indian cities

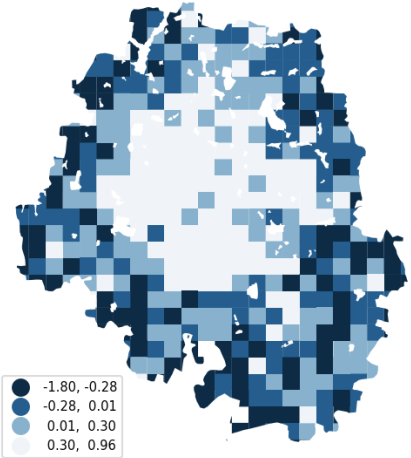
- Synthetic aperture radar to measure flooding
  - Impervious to weather, works at night
  - Define flooding as deviation from norm in surface water extent
- Can incorporate more granular satellite data (Patel 2024)
  - With customization for urban environments
- Today: data for Bangalore
  - Relative Wealth Index from Meta Data for Good
  - Stormwater drains from OpenCity India

# Flooding and wealth

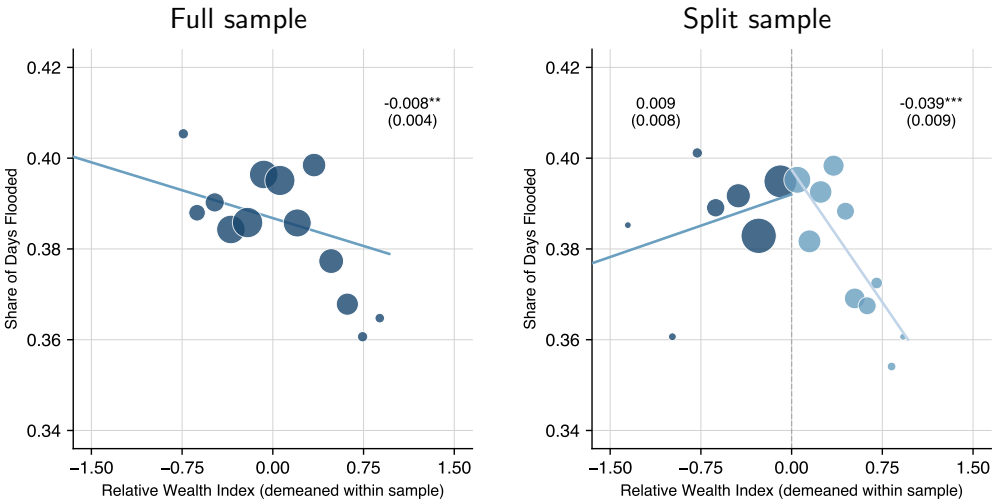
Share of days flooded



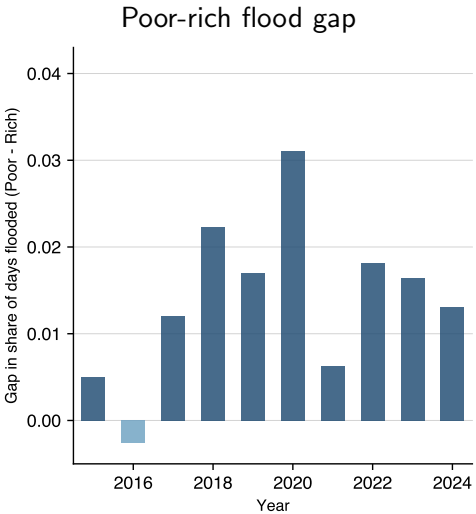

Relative wealth index

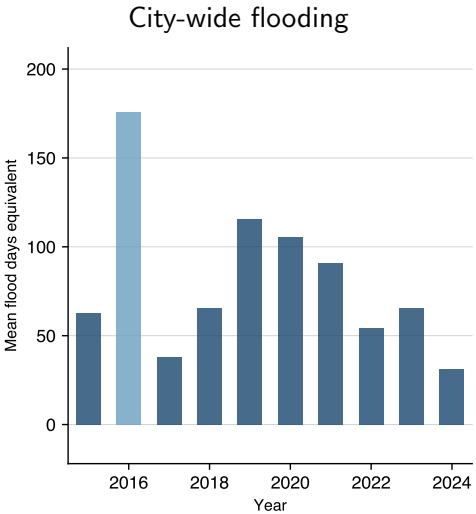


# Flooding and wealth across space



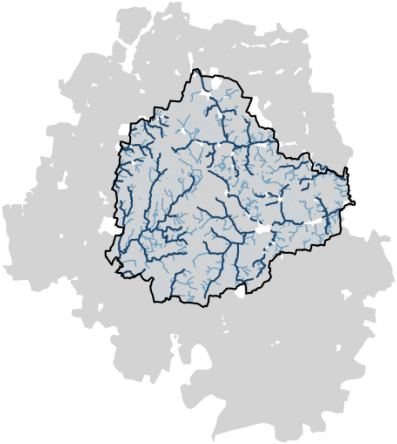
# Flooding and wealth over time



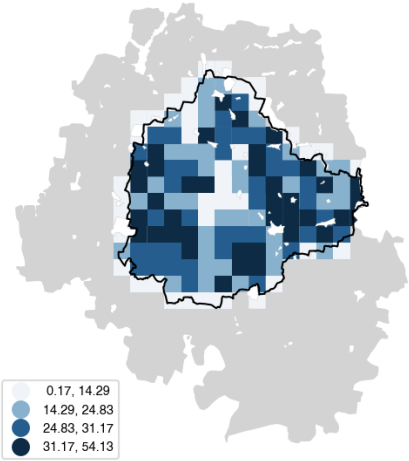



# Infrastructure

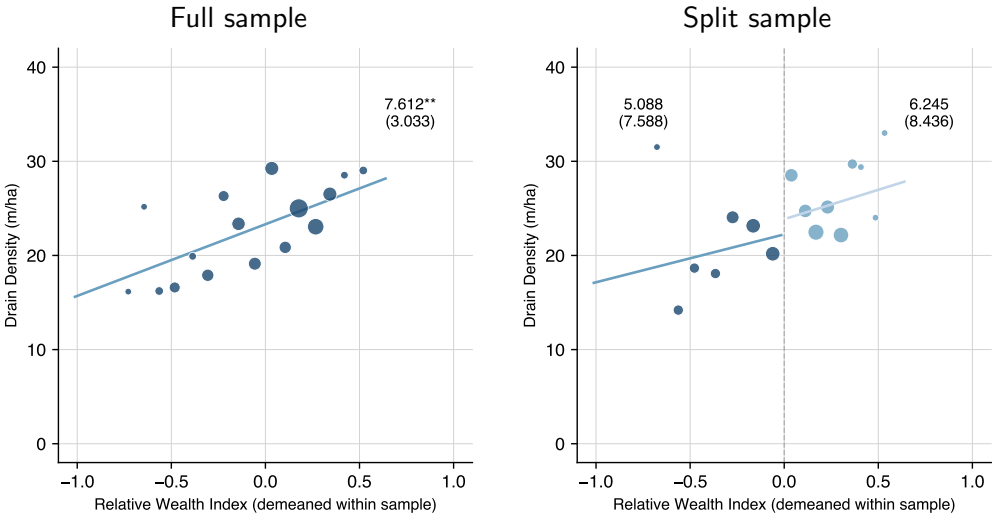
Stormwater drains



Density



# Infrastructure and wealth across space



# Next steps

- We need more work on adaptation in developing countries
  - Urban flooding, extreme heat, air pollution, sea level rise
- Adaptation policy faces trade-offs